Notation

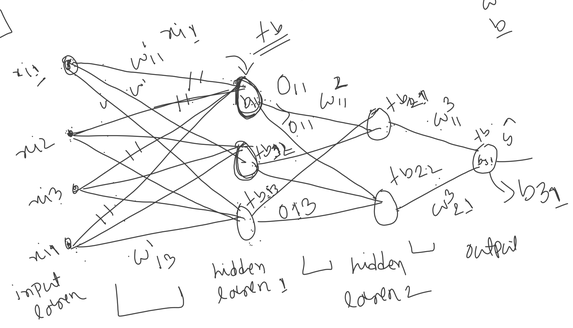
## Notation

$b_{11}$ $b_{21}$
$b_{12}$ $b_{22}$
$b_{13}$

$w$
$b$

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|

$w$ $b$




## dummy dataset

| Study hours | Sleep | Friend | Attendance | Pass/fail |
|---|---|---|---|---|
| 2 | 3 | 4 | 5 | 0 |
| 1 | 2 | 3 | 4 | 0 |
| 5 | 6 | 7 | 8 | 1 |
| 9 | 10 | 11 | 12 | 1 |

$w^1_{11}$
$w^1_{21}$
$w^1_{31}$
$w^1_{41}$

number of parameters

$w \rightarrow$ 12 6 2 $= 20$

$b \rightarrow$ 3 2 1 $= 6$

trainable parameters: $= 26$

$\text{float16} \rightarrow$ float16
float32

## Notation

$b_{ijn} = b_{ij}$

i = কোন লেয়ার
j = সেই লেয়ারের কোন Node এর

## output

Same as $b_{ijn}$

## weight

$w^k_{ij}$

$w^1_{11}$ $w^1_{21}$ $w^1$ $w^1$
$w^1_{12}$ $w^1_{22}$ $w^1$ $w^1$
$w^1_{13}$ $w^1_{23}$ $w^1$ $w^1$

k = কোন লেয়ারের ওয়েট
i
j
i,j হচ্ছে Node wise

ex: $w^1$ লেয়ারের ওয়েট

# Forward Propagation

Forward propagation is the process of computing the output of a neural network by successively applying linear transformations and activation functions from the input layer to the output layer.

prediction: $\sigma(w^T x + b)$

$$\hat{y} = w^T \cdot x + b$$

$f(w, b)$

First layer

$$\begin{bmatrix} w^1_{11} & w^1_{12} & w^1_{13} \\ w^1_{21} & w^1_{22} & w^1_{23} \\ w^1_{31} & w^1_{32} & w^1_{33} \\ w^1_{41} & w^1_{42} & w^1_{43} \end{bmatrix}^T \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

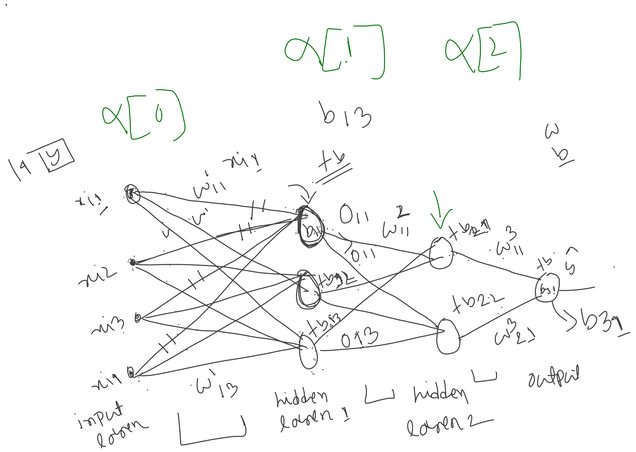
$$= \begin{bmatrix} w^1_{11} & w^1_{21} & w^1_{31} & w^1_{41} \\ w^1_{12} & w^1_{22} & w^1_{32} & w^1_{42} \\ w^1_{13} & w^1_{23} & w^1_{33} & w^1_{43} \end{bmatrix} \cdot \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ x_{i4} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} w_{11} \cdot x_1 + w_{21} \cdot x_2 + w_{31} \cdot x_3 + w_{41} \cdot x_4 \\ w_{12} \cdot x_1 + w_{22} \cdot x_2 + w_{32} \cdot x_3 + w_{42} \cdot x_4 \\ w_{13} \cdot x_1 + w_{23} \cdot x_2 + w_{33} \cdot x_3 + w_{43} \cdot x_4 \end{bmatrix}$$

$$= \sigma \left( \begin{bmatrix} w_{11} \cdot x_1 + w_{21} \cdot x_2 + w_{31} \cdot x_3 + w_{41} \cdot x_4 + b_{11} \\ w_{12} \cdot x_1 + w_{22} \cdot x_2 + w_{32} \cdot x_3 + w_{42} \cdot x_4 + b_{12} \\ w_{13} \cdot x_1 + w_{23} \cdot x_2 + w_{33} \cdot x_3 + w_{43} \cdot x_4 \end{bmatrix} \right)$$

$$= \begin{bmatrix} o_{11} \\ o_{12} \\ o_{13} \end{bmatrix}$$

## Layer 02




$$\alpha[1] = \sigma\left(\alpha[0] \cdot W^{[1]} + b^{[1]}\right)$$

$$= \alpha[2] = \sigma\left(\alpha[1] \cdot W^{[2]} + b^{[2]}\right)$$

$$\alpha[3] = \sigma\left(\alpha[2] \cdot W^{[3]} + b^{[3]}\right)$$

$$= \begin{bmatrix} w_{11}^2 & w_{12}^2 \\ w_{21}^2 & w_{22}^2 \\ w_{31}^2 & w_{32} \end{bmatrix}^T \begin{bmatrix} o_{11} \\ o_{12} \\ o_{13} \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$= \underset{1\times 3}{\begin{bmatrix} w_{11}^2 & w_{21}^2 & w_{31}^2 \\ w_{12}^2 & w_{22}^2 & w_{32}^2 \end{bmatrix}} \underset{3\times 1}{\begin{bmatrix} o_{11} \\ o_{12} \\ o_{13} \end{bmatrix}} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$= \sigma\left(\begin{bmatrix} w_{11}^2 \cdot o_{11} + w_{21}^2 \cdot o_{12} + w_{31}^2 \cdot o_{13} + b_{21} \\ w_{12}^2 \cdot o_{11} + w_{22}^2 \cdot o_{12} + w_{32}^2 \cdot o_{13} + b_{22} \end{bmatrix}\right)$$

$$= \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix}$$

Layer 3:

$$\begin{bmatrix} w^3_{11} \\ w^3_{21} \end{bmatrix}^T \cdot \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix} + b_{31}$$

$$= \begin{bmatrix} w^3_{11} & w^3_{21} \end{bmatrix} \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix} + b_{31}$$

$$= \sigma\left(w^3_{11} \cdot o_{21} + w^3_{21} \cdot o_{22} + b_{31}\right)$$

$$= \hat{y}_1 = o_{31}$$

Let's simplify: